

Step 3 : BCNF and Anomalies

Notes: In Fine, no non-key attribute determines another. reason does not determine amount, date assessed determines nothing. Rate-and-fee rule links amount to due-date, but due-date is in loan, so it's not an FD of this relation.

- 1) loan_id is NULL for damage/loss fines, so dependency isn't total over relation.
- 2) for WishlistItem: isbn_issn is a candidate key, along with wish_id, because the wishlist is staff-maintained, one row per title.

Method: for each relation, list non-trivial FDs, find candidate keys, check every FD has a superkey on the left. Decompose if not.

Assumptions about the domain:

The FDs below depend on how our library operates, so we state these rules first. Each one either creates an FD or rules one out.

- **A1.** A person is identified by person_id. Email addresses are unique, so no two people share one.
- **A2.** A library card number is unique and belongs to exactly one member.
- **A3.** A barcode is printed on one physical copy and never reused.
- **A4.** An ISBN identifies one edition of one book, so it maps to one Item.
- **A5.** An ISSN identifies a *serial series* (a magazine or journal title), not a single issue. Many issues share one ISSN.
- **A6.** The loan period depends on the material type (3 weeks for books and serials, 1 week for records and high-demand items) and staff may override it. Renewals extend the due date and are counted in renewal_count.
- **A7.** Salary is negotiated per individual, not fixed by role band.
- **A8.** Age suitability is decided per event, not per event type. The same kind of event can be all-ages one week and adults-only the next.
- **A9.** A room cannot host two events at the same time.
- **A10.** shelf_location is a full call number assigned per title, depending on author and title as well as subject, so subject alone does not determine it. All copies of a title are shelved together, so the location is a property of the Item, not of each Copy.
- **A11.** Fines may be charged for damage or loss with no overdue loan attached, so loan_id is NULL on those rows. One loan can also produce more than one fine (for example overdue *and* damaged).
- **A12.** The wishlist is maintained by staff as an acquisitions list, with one row per title. A title whose details are not yet confirmed has a NULL isbn_issn.

Analysis:

Relations are grouped in the same order as the Step 2 diagrams.

Items and copies

Item(item_id, title, pub_year, subject, language, shelf_location)

- FDs: item_id → all others
- Keys: item_id
- BCNF. Two items can share a title, by A10 subject → shelf_location does not hold.

Book(item_id, isbn, author, publisher, pages)

- FDs: item_id → all others; isbn → all others (A4)
- Keys: item_id, isbn
- BCNF. Both left sides are candidate keys.

Serial(item_id, issn, volume, issue_no, publisher) - violates BCNF

- FDs: item_id → all others; {issn, volume, issue_no} → item_id, publisher;

issn → publisher

- Keys: item_id, {issn, volume, issue_no}

By A5 one ISSN covers many issues, so issn is not a superkey and

issn → publisher is a **bad FD**. Concretely: changing a magazine's publisher

would mean updating every issue row (update anomaly), and we could not record a

publisher for a series before its first issue arrives (insert anomaly).

Decompose on $X = \{\text{issn}\}$, where $X^+ = \{\text{issn}, \text{publisher}\}$:

$R_1 = X^+ = \text{SerialSeries}(\text{issn}, \text{publisher})$

$R_2 = X \cup \{\text{rest}\} = \text{Serial}(\text{item_id}, \text{issn}, \text{volume}, \text{issue_no})$

SerialSeries(issn, publisher) - FD issn → publisher; key issn. BCNF.

Serial(item_id, issn, volume, issue_no) - FDs item_id → all others and {issn, volume, issue_no} → item_id; keys item_id, {issn, volume, issue_no}.

BCNF. issn → publisher no longer applies, since publisher has left.

The decomposition is **lossless** (issn is the key of SerialSeries) and

preserves dependencies.

Record(item_id, artist, label, runtime_min, genre)

- FDs: item_id → all others
- Keys: item_id
- BCNF. artist → label does not hold; artists release on different labels.

Copy(copy_id, item_id, barcode, condition, acquired_date)

- FDs: copy_id -> all others; barcode -> all others (A3)
- Keys: copy_id, barcode
- BCNF. Note item_id -> shelf_location *would* be a bad FD here, since many copies share one item. We avoid it by keeping shelf_location in Item (A10).

People

Person(person_id, first_name, last_name, email, phone, date_of_birth, address)

- FDs: person_id -> all others; email -> all others (A1)
- Keys: person_id, email
- BCNF. Two people can share an address, so address determines nothing.

Member(person_id, card_number, member_since, card_status)

- FDs: person_id -> all others; card_number -> all others (A2)
- Keys: person_id, card_number
- BCNF.

Staff(person_id, role, hire_date, salary, supervisor_id)

- FDs: person_id -> all others
- Keys: person_id
- BCNF. By A7 role -> salary does not hold; if it did it would be a bad FD and would force a SalaryBand(role, salary) relation.

Circulation

Loan(loan_id, copy_id, member_id, borrow_date, due_date, renewal_count, return_date)

- FDs: loan_id -> all others
- Keys: loan_id
- BCNF. borrow_date -> due_date would be a bad FD (many loans start the same day), but by A6 the period varies and renewals push the due date back, so it does not hold. {copy_id, borrow_date} is not a key either: a copy returned in the morning can be lent again the same afternoon.

Fine(fine_id, member_id, loan_id, reason, amount, date_assessed, amount_paid, date_paid)

- FDs: fine_id -> all others
- Keys: fine_id
- BCNF. loan_id is not a key (A11: one loan can yield an overdue fine and a damage fine), so loan_id -> member_id would be a bad FD — but it is not total, since loan_id is NULL on damage and loss fines. Where it is present, a trigger enforces agreement with Loan.member_id (Step 4). Dropping member_id instead would leave non-loan fines with no owner. No non-key

attribute determines another: reason does not determine amount, and the rate-and-cap rule links amount to due_date, which is in Loan, not here.

Events

Room(room_id, name, room_type, floor, capacity)

- FDs: room_id -> all others; name -> all others
- Keys: room_id, name
- BCNF. room_type -> capacity does not hold; two meeting rooms can seat different numbers.

Event(event_id, title, description, event_type, event_date, start_time, end_time, min_age, max_age, max_attendees, room_id, organizer_id)

- FDs: event_id -> all others; {room_id, event_date, start_time} -> all others (A9)
- Keys: event_id, {room_id, event_date, start_time}
- BCNF. By A8 event_type -> min_age, max_age does not hold. Nor does room_id -> max_attendees: the cap is chosen per event and may sit below the room's physical capacity, which lives in Room.

EventRegistration(event_id, member_id, date_registered, attended)

- FDs: {event_id, member_id} -> date_registered, attended
- Keys: {event_id, member_id}
- BCNF. A member registers for an event at most once; neither non-key attribute determines anything.

Services and acquisitions

HelpRequest(request_id, member_id, staff_id, question, date_asked, response, date_answered, status)

- FDs: request_id -> all others
- Keys: request_id
- BCNF. status -> date_answered does not hold; many requests share the status 'Answered' but were answered on different days.

WishlistItem(wish_id, title, creator, item_type, isbn_issn, est_cost, priority, date_requested, status, acquired_item)

- FDs: wish_id -> all others; isbn_issn -> title, creator, item_type (partial)
- Keys: wish_id, isbn_issn
- BCNF. By A12 there is one row per title, so isbn_issn is a candidate key and the second FD has a superkey on its left. Rows with unconfirmed details carry a NULL isbn_issn, so that FD is not total. status -> acquired_item does not hold; many acquired rows point at different items.

Dependencies considered and rejected

Candidate FD	Relation	Why it does not hold
role -> salary	Staff	A7: salary set per individual
borrow_date -> due_date	Loan	A6: period varies; renewals extend it
event_type -> min_age, max_age	Event	A8: age range set per event
subject -> shelf_location	Item	A10: call number also depends on author and title
room_type -> capacity	Room	same type, different seating
room_id -> max_attendees	Event	cap is per event, not per room
artist -> label	Record	artists release on different labels
status -> date_answered	HelpRequest	same status, different answer dates
status -> acquired_item	WishlistItem	many acquired rows, different items
loan_id -> member_id	Fine	A11: not total, NULL on damage and loss fines

Concl: After decomposing `Serial`, every relation has a superkey on the left of every non-trivial FD, so update, insert and delete anomalies cannot occur. The one dependency we kept, `loan_id -> member_id` in `Fine`, is not total and is enforced by trigger instead.

Final schema after decomposition (16 relations)

Item(item_id, title, pub_year, subject, language, shelf_location)

Book(item_id, isbn, author, publisher, pages)

Serial(item_id, issn, volume, issue_no)

SerialSeries(issn, publisher)

Record(item_id, artist, label, runtime_min, genre)

Copy(copy_id, item_id, barcode, condition, acquired_date)

Person(person_id, first_name, last_name, email, phone, date_of_birth, address)

Member(person_id, card_number, member_since, card_status)

Staff(person_id, role, hire_date, salary, supervisor_id)

Loan(loan_id, copy_id, member_id, borrow_date, due_date, renewal_count, return_date)

Fine(fine_id, member_id, loan_id, reason, amount, date_assessed, amount_paid, date_paid)

Room(room_id, name, room_type, floor, capacity)

Event(event_id, title, description, event_type, event_date, start_time, end_time, min_age, max_age, max_attendees, room_id, organizer_id)

EventRegistration(event_id, member_id, date_registered, attended)

HelpRequest(request_id, member_id, staff_id, question, date_asked, response, date_answered, status)

WishlistItem(wish_id, title, creator, item_type, isbn_issn, est_cost, priority, date_requested, status, acquired_item)